

CLAUDE.md — Project Rules

Project: Better Day (mobile puzzle game)

Who's involved

- **Human:** Non-developer, game designer and project owner. Uses iPhone. Testing on mobile is critical.
- **Claude (chat):** Design, prototyping, strategy, game design iteration. Single HTML/JS prototypes.
- **Claude Code:** Production development. Capacitor wrapping, persistent storage, app submission.

How we work

- Human is not a coder — explain decisions in plain English
- Always test on mobile (iPhone Safari) — desktop-only is useless
- Single HTML files with embedded JS for prototypes
- Three.js r128 for 3D rendering — confirmed working
- Prototype first before committing to approaches
- Update the 4 project files at the end of each session
- Do clean full rebuilds rather than accumulating patches
- Keep simplifying — complexity is the enemy

Critical constraints

- 3D scenes from Skybox AI (equirectangular panoramas, 2:1 ratio)
- Mosaic Mode needs only ONE image (good day) — bad day generated in software
- Image Mode needs TWO images (bad day + good day matched pair)
- Must work on iOS and Android
- Automated scene generation pipeline for production

- Monetisation built in from the start

Current prototype

- Default images embedded as base64 (ink drawing = bad day, Dutch Masters = good day)
- Upload buttons available to swap images
- File is ~460KB due to embedded images
- Two game modes: Mosaic (default) and Image

Quality checks before delivering

- Works on mobile?
- Image right way up? ($v = \text{lat}/\text{PI} + 0.5$)
- Hexagons uniform across sphere?
- Long-press vs swipe works reliably?
- Self-contained file (only CDN Three.js)?
- Preview content matches game logic? (render-to-texture only)
- Cancel restores state properly?
- No stuck states? (last tile auto-completes)
- Pole tiles excluded from gameplay?
- Inside/outside view switch works correctly?
- Controls feel natural in both views? (horizontal reverses, vertical same)
- Green/amber pulse feedback working?
- Camera centers on placed tile after switch back?

What NOT to attempt (proven failures)

- Manual UV sampling for tile previews
- Rendering foreign content on a tile's 3D mesh
- 2D overlay tracking 3D tile position
- WebGL lines for thick borders (always 1px — use tile tinting instead)

- Chain resolution (unnecessary — simple swap works)
- Mini-hexagon mosaic (hexagons don't evenly subdivide into hexagons)